

BONSEO KOO

Undergraduate Researcher in Electronic Engineering

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Research Interests: FPGA & Reconfigurable Computing · AI Accelerator Architecture · Low-Precision and Sparse Inference · Digital VLSI · HW/SW Co-Design

EDUCATION

Sogang University — B.S. Candidate in Electronic Engineering

Expected Aug. 2027

Six semesters completed | GPA: **3.79/4.50** | Official percentile: **92.4%**

- Selected coursework: Digital Circuit Design (A+), Digital Logic Circuit Laboratory (A0), Computer Architecture (A0), Advanced Digital Circuit Design (A0), Microprocessor, Probability and Random Variables (A0).

RESEARCH EXPERIENCE

Undergraduate Research Intern — Yonsei University, Department of Electrical & Electronic Engineering

Summer 2026

Advisor: Prof. Seungkyu Choi

- Completed a LeNet-based MNIST inference NPU design project on a Zynq SoC, extending experience from PL-only accelerator design toward PS-PL system integration.
- Developed PS-side control in Vitis and worked with AXI4-Lite-based memory-mapped control/data access and DMA-based data transfer for FPGA accelerator integration.
- Completed a research-paper presentation on “OliVe: Accelerating Large Language Models via Hardware-friendly Outlier-Victim Pair Quantization,” analyzing low-bit quantization, outlier handling, accelerator performance, and energy efficiency.

FPGA NPU Architecture Design & Optimization — Sogang University

Dec. 2025 – Jul. 2026

Advisor: Prof. Sungju Ryu

- Designed and implemented an FPGA NPU in Verilog RTL based on an 8×8 weight-stationary systolic array and matrix-tiling controller; validated the design at 100 MHz on PYNQ-Z2.
- Proposed LOD-based Dynamic Scaling and Switching-Aware Zero-Skipping to improve the fidelity–area–power trade-off of low-precision inference; six directed testbenches collectively exercised all 64 supported sparsity maps.
- Evaluated 45 NPU configurations; the selected design achieved 92.0% fidelity versus 74.0% for a fixed 7-bit baseline while reducing LUT usage by 23.7%, BRAM usage by 11.1%, and dynamic power by 12.9%.
- Integrated the FPGA NPU into a live-camera CIFAR-10 inference pipeline and demonstrated the system at the 2026 ISE Summer Conference.

PUBLICATION & CONFERENCE PRESENTATION

B. Koo, Y. Lee, T. Jung, and S. Ryu, “Maximizing NPU Fidelity and Efficiency with LOD-Based Dynamic Scaling and Switching-Aware Zero-Skipping,” *2026 ISE Summer Conference*, first-author oral presentation, Jul. 2026.

AWARDS

Live Demonstration Excellence Award — 2026 ISE Summer Conference

Jul. 15, 2026

- Awarded for the live FPGA demonstration of the low-precision NPU and CIFAR-10 inference pipeline.

ONGOING ENGINEERING RESEARCH

End-to-End CIFAR-10 FPGA CNN Accelerator — Sogang University

Aug. 2026 – Present

- Implemented and validated an end-to-end SmallVGG inference pipeline on PYNQ-Z2, covering Conv1–Conv6, global average pooling, and FC1–FC2 with Vitis-based Zynq PS-PL control.
- Achieved 91.5% accuracy on a 1,000-image CIFAR-10 evaluation versus 91.9% for the software reference using integer shift-and-round requantization; validated matching FPGA and reference logits in end-to-end testing.
- Currently implementing channel-chunked buffering and line-buffer-based streaming window generation to reduce intermediate feature-map movement; evaluating resource, Fmax, cycle, latency, power, and energy trade-offs against the full-buffer baseline.

TECHNICAL SKILLS

Hardware / RTL: Verilog HDL, RTL Design, FPGA Design, Systolic Arrays, NPU Architecture, Fixed-Point Arithmetic, RTL Simulation & Verification

FPGA / SoC: AMD/Xilinx Vivado, Vitis, PYNQ-Z2, Zynq PS-PL Integration, AXI4-Lite, DMA, BRAM

Programming / ML: Python, PyTorch, Jupyter Notebook, C

Other Tools: MATLAB, Cadence Virtuoso, PSpice, TCAD, Visual Studio

LANGUAGE

TOEFL iBT: 5.0/6.0 overall (comparable to 100/120; CEFR C1)